NAME ____________________ DATE ______________

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1 Physiology and levels of function

COMP

You should be able to: Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

PICK ONE AND DO THAT ONE

- [ ] **A** Build your own ladder. Pick a process, not oxygen delivery, and draw it at all six levels, molecule up to organism. One small box per level, labeled. Then break one rung and write, level by level, what stops working above it.
- [ ] **B** Pick your own process, not oxygen delivery. Draw the same thing twice. First as anatomy sees it, the structures and their parts. Then as physiology sees it, the same thing working, with arrows for what moves and what changes. Under each drawing, one line naming the question it answers.

PASS 1 COLOR __________ PASS 2 COLOR __________

DRAW IT

split it up however the prompt needs

2 Structure and function relationship

COMP

You should be able to: Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one structure from this week and label the single feature that makes its job possible. Then sketch what would stop working if that feature changed.
- [ ] **B** List three structures from this week. Beside each, write the one property its function depends on, in six words or fewer.

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3 Homeostasis defined

COMP

You should be able to: Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

PICK ONE AND DO THAT ONE

- ☐ **A** Write the definition of homeostasis, then underline the three words doing the real work. Beside each underlined word, one line on why it matters.
- ☐ **B** Draw two graphs side by side, a value in steady state and a value at equilibrium. On each, label what is still moving and what it costs.

PASS 1 COLOR ____________ PASS 2 COLOR ____________

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4 Feedback loop components

COMP

You should be able to: Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the five component reflex pathway top down and label all five plus the feedback arrow. Labels only, no sentences.
- ☐ **B** Fill in the five components for thermoregulation on a cold day. Then mark which component you would suspect first if the patient never shivered.

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5 Negative and positive feedback

COMP

You should be able to: Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one negative feedback loop and one positive feedback loop as arrow diagrams. On each, mark what makes it stop.
- ☐ **B** Draw clotting as a positive feedback loop, then show what ends it. Underneath, write one line on why the body can afford to use positive feedback here.

PASS 1 COLOR ____________ PASS 2 COLOR ____________

DRAW IT

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6 Feedforward control and acclimatization

COMP

You should be able to: Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a timeline of eating a meal. Mark where feedforward starts and where feedback takes over.
- ☐ **B** Draw a setpoint as a horizontal line. Show what fever does to it, and mark at each stage what the patient feels.

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7

COMP

Local and reflex control pathways

You should be able to: Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a local control example and a reflex control example side by side. On each, mark how far the signal travels.
- ☐ **B** Standing up suddenly: draw the reflex, label all five components, and write one line on why local control could not have handled it.

PASS 1 COLOR ____________ PASS 2 COLOR ____________

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split it up however the prompt needs

8

COMP

Mass balance

You should be able to: Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

PICK ONE AND DO THAT ONE

- ☐ **A** Write the mass balance equation and label every term. Then draw a body outline with arrows in and out for sodium, labeling which arrow is which term.
- ☐ **B** Work a mass balance for one substance across one day. Then change a single term and show the new result underneath.

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9 Units and unit conversion

COMP

You should be able to: Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

PICK ONE AND DO THAT ONE

- [ ] **A** Take one real lab value and walk it through every unit physiology reports it in. Glucose in mg/dL to mmol/L, or a pressure in mmHg to kPa. Show the conversion as a drawn chain, one box per step, with the factor written on the arrow between them. Then write one line on which unit a clinician would actually say out loud, and why that one.
- [ ] **B** Draw a number line for one quantity, say volume, and mark liters, milliliters and microlitres on it in the right places. Put three real physiological volumes on that line: a tidal breath, a drop of blood taken for a glucose stick, and the water in an adult body. Under it, write what goes wrong in a patient if you read one of them off by a factor of a thousand.

PASS 1 COLOR ____________ PASS 2 COLOR ____________ ALSO A LAB SKILL

DRAW IT

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10 Graphing and data interpretation

COMP

You should be able to: Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

PICK ONE AND DO THAT ONE

- [ ] **A** Plot the mission patient's sodium across the three days. Label both axes with units and mark the reference range on the graph.
- [ ] **B** Draw a saturation curve. Label the plateau, then write one line on what the plateau means physiologically.

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11 Experimental design and controls

COMP

You should be able to: Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

PICK ONE AND DO THAT ONE

- [ ] **A** Sketch an experiment as three labeled boxes: what you changed, what you measured, and your control.
- [ ] **B** Take a claim from this week and design the control group that would test it. Draw both groups and label the one difference between them.

PASS 1 COLOR ____________ PASS 2 COLOR ____________ ALSO A LAB SKILL

DRAW IT

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12 Measurement error and variability

COMP

You should be able to: Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw two sets of repeated measurements, one with high variability and one with low. Mark which difference between readings is real.
- [ ] **B** List three sources of variation in a single measurement. Beside each, one way to reduce it.

PASS 1 COLOR ____________ PASS 2 COLOR ____________ ALSO A LAB SKILL

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